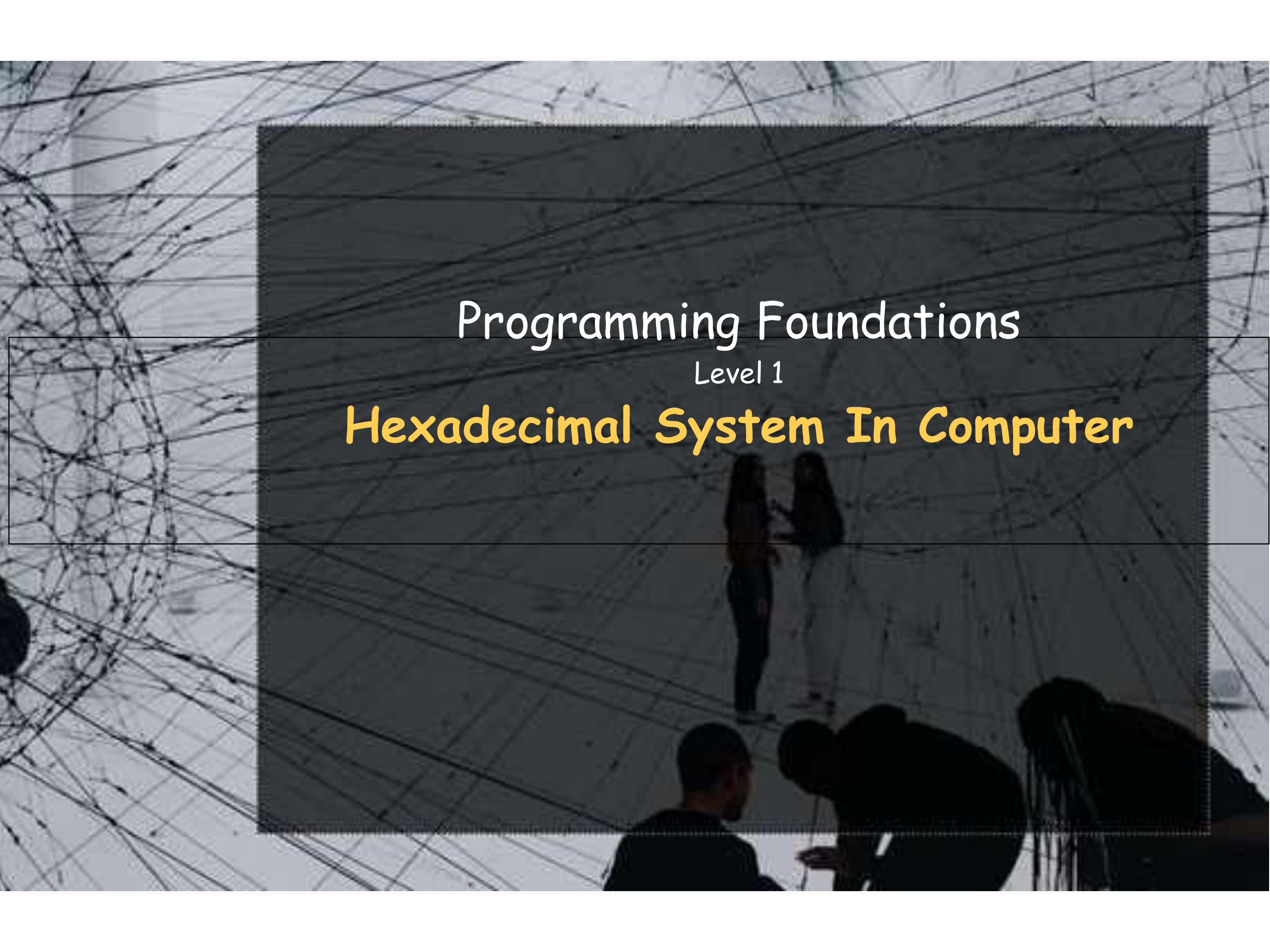




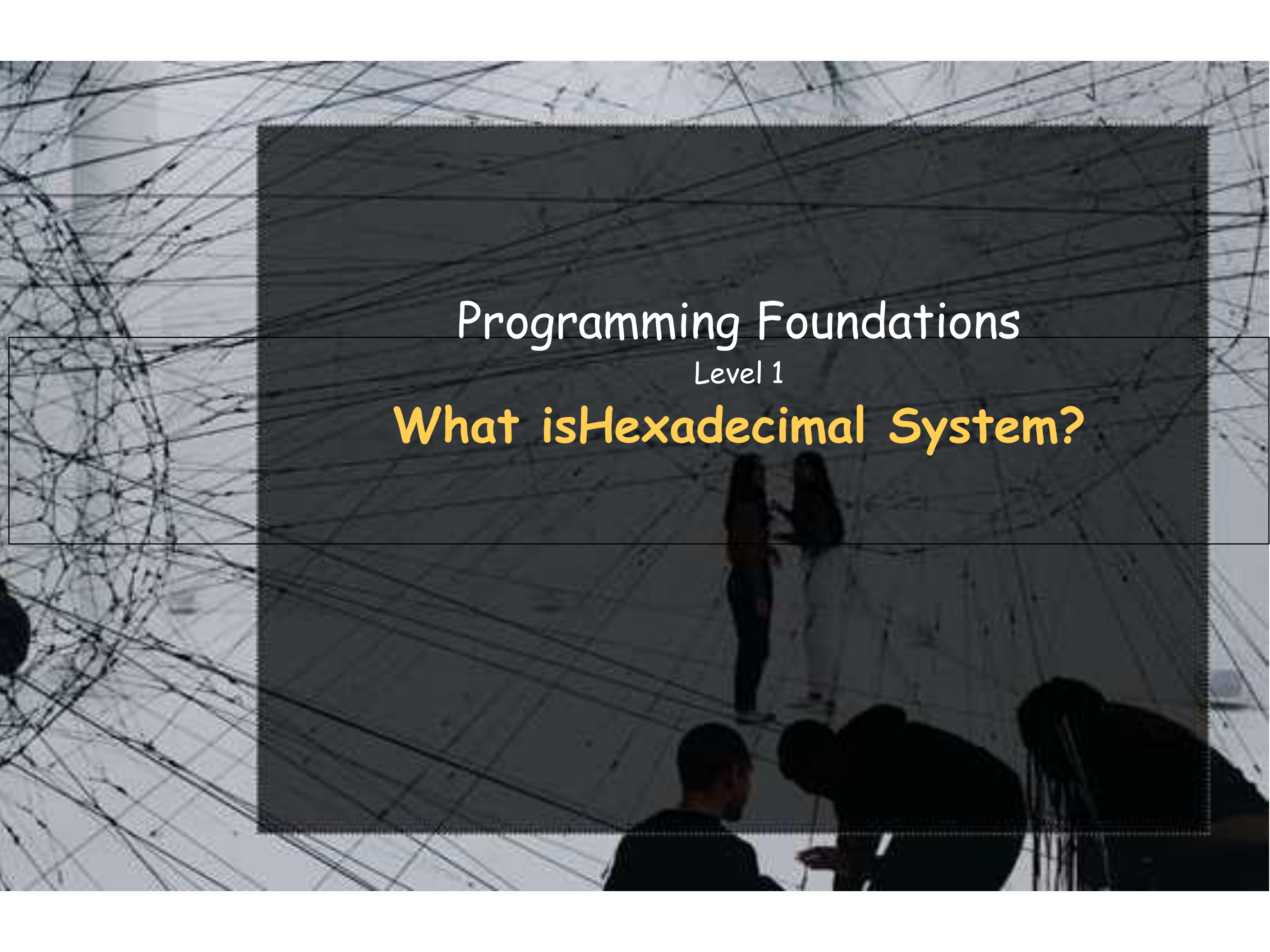
Programming Foundations Level 1



Programming Foundations

Level 1

Hexadecimal System In Computer



Programming Foundations

Level 1

What is Hexadecimal System?

What Is the Hexadecimal System (Base-16)?

Computers speak binary (base-2), but long strings of 0/1 are hard for humans to read. Hexadecimal (base-16) gives a compact, human-friendly way to view the same bits: every 4 binary bits (a nibble) map to one hex digit. One byte (8 bits) maps neatly to 2 hex digits — perfect for memory dumps, colors, Unicode, and machine code

Decimal vs Binary vs Hexadecimal?



What Is the Hexadecimal System (Base-16)?

- Decimal (base-10): digits 0-9
- Binary (base-2): digits 0-1
- Hexadecimal (base-16): digits 0-9 + A, B, C, D, E, F (A=10 ... F=15)

Key 4-bit mapping:

Decimal

Binary

Hexadecimal

0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

Why Hexadecimal?

It's hard to human to read binary!

What does this mean ?

01001001 00100000 01001100 01101111 01110110 01100101 00100000
01011001 01101111 01110101 00100001



I Love You!



Hexadecimal System

Provides a **human-friendly representation**

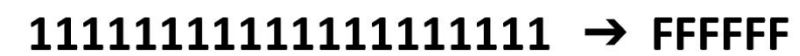
11010100 → D4

Each Byte is represented with 2 letters only.

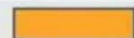
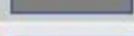
What Is the Hexadecimal System (Base-16)?

- Humans struggle with long binary strings (e.g., "I Love You!" in ASCII bits).
- Hex compresses binary:
 - 11010100 → **D4**
 - 1111 1111 1111 1111 1111 → **FFFFFF** (grouped by 4 bits)
- **Each byte = 2 hex digits** → clean, readable dumps and color codes

- **11010100 → D4**
- **1111 1111 1111 1111 1111 → FFFFFFFF**
- Each Byte is represented with 2 letters only.



Examples:

Color	CSS Color Name	Hex Code #RPGGBB	Decimal Code (R,G,B)
	Red	#FF0000	rgb (255,0,0)
	Orange	#FFA500	rgb (255,165,0)
	Yellow	#FFFF00	rgb (255,225,0)
	Green	#008000	rgb (0,128,0)
	Cyan	#00FFFF	rgb (0,255,225)
	Blue	#0000FF	rgb (0,0,225)
	Purple	#800080	rgb (128,0,128)
	Pink	#FFC0CB	rgb (255,192,203)
	Gray	#808080	rgb (128,128,128)
	Brown	#A52A2A	rgb (165,42,42)

The background of the slide features a complex, abstract pattern of thin, dark lines crisscrossing over a lighter, textured surface. In the lower portion of the image, the silhouettes of several people are visible, appearing to be in a collaborative setting, possibly a meeting or a workshop, with some individuals leaning over a table or screen.

Programming Foundations

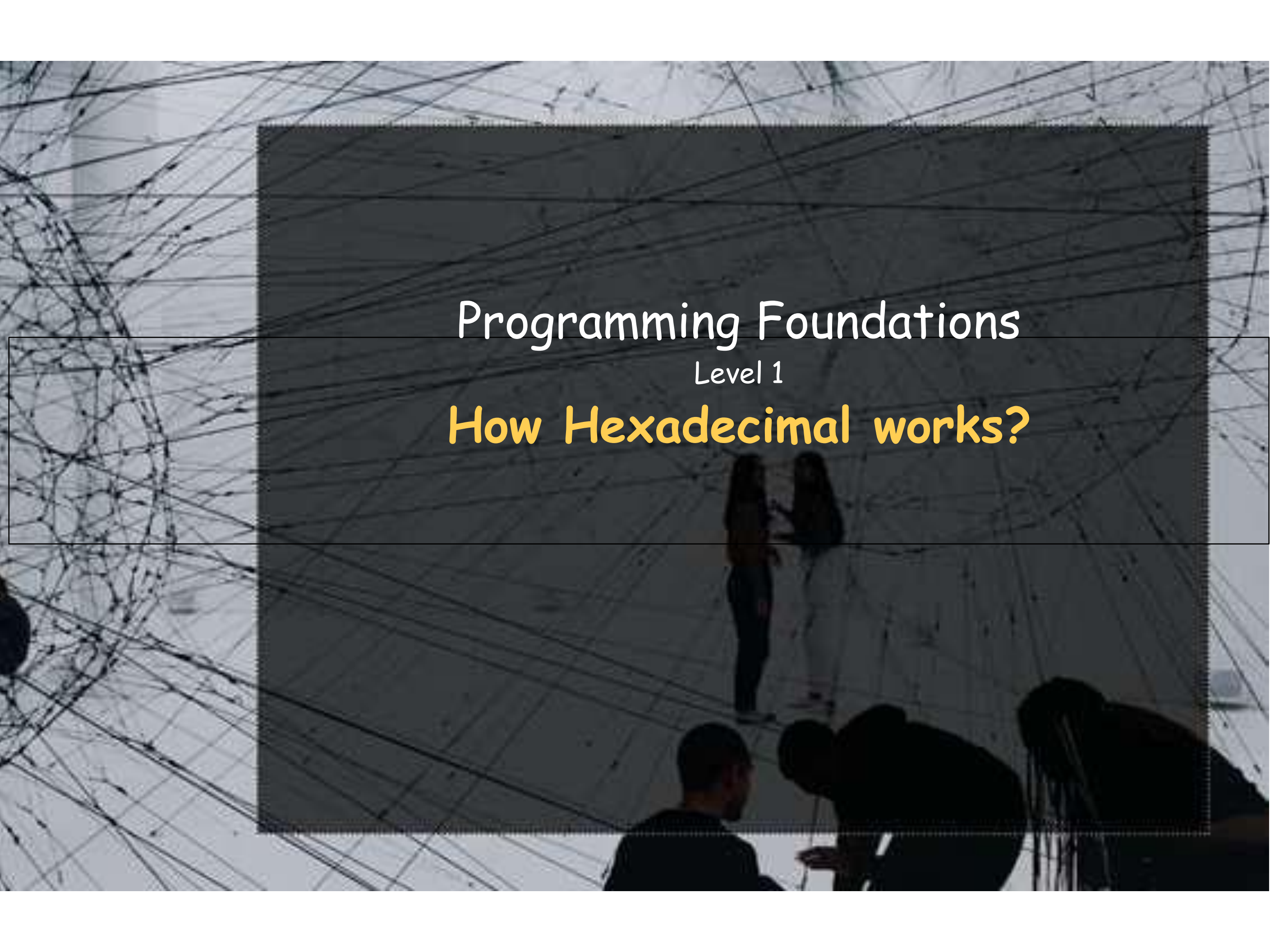
Level 1

Hexadecimal Prefex

Prefix:

<u>Technology/Language</u>	<u>Prefix</u>	<u>Example</u>
HTML & CSS	#Code	#FFFFFF
C,C++,Java..etc	0xCode	0xC2A4
XML	&#Code	&#C2A4
Unicode	U+Code	U+C2A4

Tech/Language	Prefix	Example
HTML/CSS	#	#FFFFFF
C / C++ / Java...	0x	0xC2A4
XML entities	&#	&#C2A4
Unicode	U+	U+C2A4



Programming Foundations

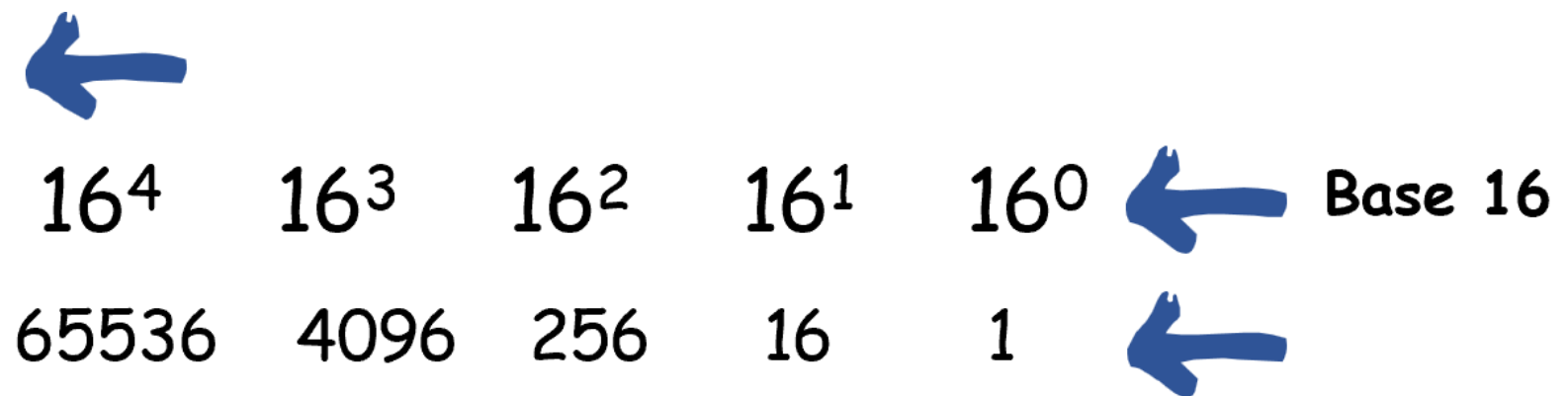
Level 1

How Hexadecimal works?

Remember that binary was base 2.

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0		Base 2
128	64	32	16	8	4	2	1		Doubling

Remember that binary was base 2.



The diagram illustrates the powers of 16 and their corresponding decimal values. It consists of two rows of numbers. The top row shows powers of 16: 16^4 , 16^3 , 16^2 , 16^1 , and 16^0 . The bottom row shows the decimal equivalents: 65536, 4096, 256, 16, and 1. To the right of the top row is the text 'Base 16'. There are three blue arrows pointing to the left: one above the 16^4 column, one to the right of the 16^0 column, and one below the 1 column.

16^4	16^3	16^2	16^1	16^0	Base 16
65536	4096	256	16	1	



Programming Foundations

Level 1

Convert Decimal to Hex

Remember...

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

Convert 469 to Hex

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

<u>Number / 16</u>	<u>Result</u>	<u>Integer</u>	<u>Fraction</u>	<u>Remainder</u>	
469 / 16 =	29.3125	29	0.3125	16 × 0.3125 =	5 →
29 / 16 =	1.8125	1	0.08125	16 × 0.08125 =	13 →
1/16 =	--	--	--		1 →

↑

$469_{10} \rightarrow 1D5_{16}$

Convert 1513 to Hex

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

<u>Number / 16</u>	<u>Result</u>	<u>Integer</u>	<u>Fraction</u>	<u>Remainder</u>		<u>Hex</u>
1513 / 16 =	94.5625	94	0.5625	16 × 0.5625 =	9	→ 9
94 / 16 =	5.875	5	0.875	16 × 0.875 =	14	→ E
5/16 =	--	--	--		5	→ 5

1513₁₀ → 5E9₁₆

Convert 479 to Hex

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

<u>Number / 16</u>	<u>Result</u>	<u>Integer</u>	<u>Fraction</u>	<u>Remainder</u>		<u>Hex</u>
479 / 16 =	29.9375	29	0.9375	16 × 0.9375 =	15 →	F
29 / 16 =	1.8125	1	0.8125	16 × 0.8125 =	13 →	D
1/16 =	--	--	--		1 →	1

$479_{10} \rightarrow 1DF_{16}$



Programming Foundations

Level 1

Convert Hex to Decimal

Convert **1D5** to Decimal

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

↑

16^2 16^1 16^0
↖ ↗
1D5

$$5 \times 16^0 = 5 \times 1 = 5 +$$

$$D \times 16^1 = 13 \times 16 = 208 +$$

$$1 \times 16^2 = 1 \times 256 = 256$$

469

Convert **5E9** to Decimal

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
	↑														

16^2 16^1 16^0
↖ ↗
5E9

$$9 \times 16^0 = 9 \times 1 = 9 +$$

$$E \times 16^1 = 14 \times 16 = 224 +$$

$$5 \times 16^2 = 5 \times 256 = 1280$$

1513

Convert 1DF to Decimal

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
↑		↑													

16^2 16^1 16^0
↑ ↑ ↑
1DF

$$\begin{array}{rclcl} F \times 16^0 & = & 15 \times 1 & = & 15 + \\ D \times 16^1 & = & 13 \times 16 & = & 208 + \\ 1 \times 16^2 & = & 1 \times 256 & = & 256 \\ \hline & & & & 479 \end{array}$$



Programming Foundations

Level 1

Convert Hex to Binary

How to Convert Hexa to Binary?

- Two steps:
 1. Convert hexa to decimal.
 2. Convert decimal to binary.

That's it.

Convert 1D5 to Binary

- Two steps:
 1. Convert 1D5 to decimal.
 2. Convert decimal to binary.

That's it.

Step1:

Convert 1D5 to Decimal

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

↑

16^2 16^1 16^0
↖ ↗
1D5

$$5 \times 16^0 = 5 \times 1 = 5 +$$

$$D \times 16^1 = 13 \times 16 = 208 +$$

$$1 \times 16^2 = 1 \times 256 = 256$$

469

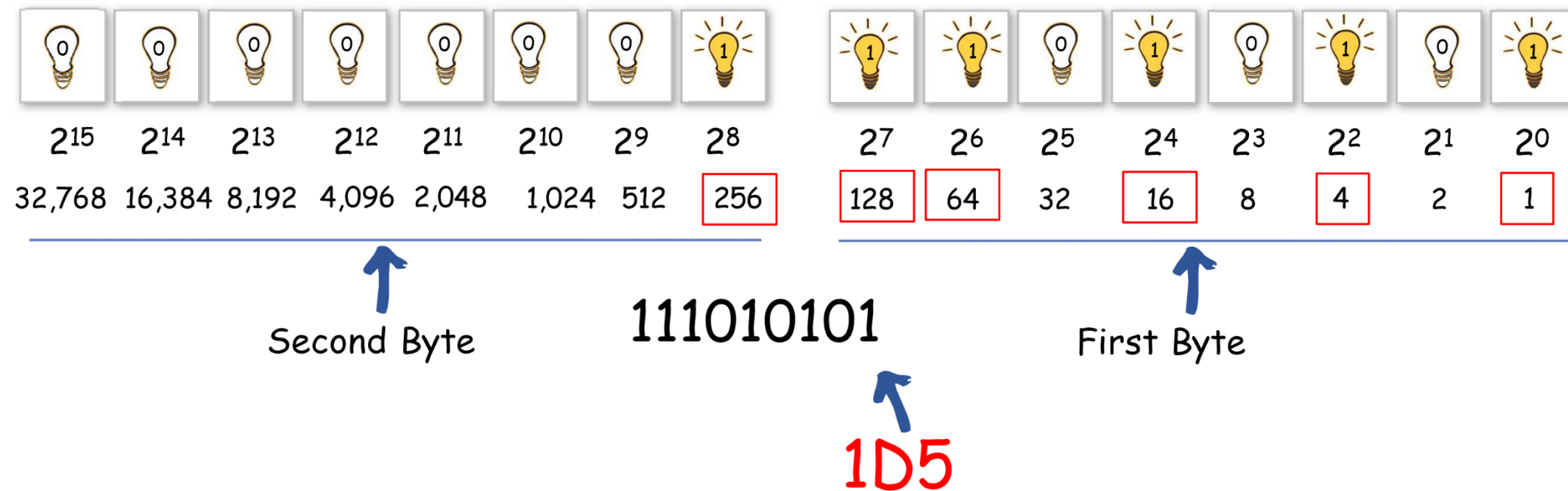


Now convert
this to binary

Step2:

Convert 469 to Binary

469





Programming Foundations

Level 1

Convert Binary to Hexa

How to Convert Hexa to Binary?

- Two steps:
 1. Convert binary to decimal.
 2. Convert decimal to hexa.

That's it.

Interconnection:

- ◆ 4 bits $\leftrightarrow$ 1 hex digit $\rightarrow$ compact human view of binary.
- ◆ 1 byte $\leftrightarrow$ 2 hex digits $\rightarrow$ easy memory/ASCII/color dumps.
- ◆ Powers of 16 (1,16,256,4096...) $\rightarrow$ drive decimal $\leftrightarrow$ hex math.
- ◆ Prefixes (#, 0x, &#, U+) $\rightarrow$ tell tools to interpret as hex.

By mastering hex, you read/write low-level data confidently — firmware, encodings, colors, addresses — without drowning in 0/1. 



Activity



Instructions:

- 1 Convert **469**, **1513**, **479** to hex (show steps).
- 2 Convert **1D5**, **5E9**, **1DF** to decimal (expand by 16^n).
- 3 Convert **13B4** and **A5** to **binary** using nibble mapping



Activity Solution



Solution:

① Convert 469, 1513, 479 to hex (show steps)

469 → Hex

$$469 \div 16 = 29, \text{ remainder } 5$$

$$29 \div 16 = 1, \text{ remainder } 13 \rightarrow \text{D}$$

$$1 \div 16 = 0, \text{ remainder } 1$$

Read remainders bottom→top → 1 D 5

$$\rightarrow 469_{10} = 1\text{D}5_{16}$$

1513 → Hex

$$1513 \div 16 = 94, \text{ remainder } 9$$

$$94 \div 16 = 5, \text{ remainder } 14 \rightarrow \text{E}$$

$$5 \div 16 = 0, \text{ remainder } 5$$

Read remainders bottom→top → 5 E 9

$$\rightarrow 1513_{10} = 5\text{E}9_{16}$$

479 → Hex

$$479 \div 16 = 29, \text{ remainder } 15 \rightarrow \text{F}$$

$$29 \div 16 = 1, \text{ remainder } 13 \rightarrow \text{D}$$

$$1 \div 16 = 0, \text{ remainder } 1$$

Read remainders bottom→top → 1 D F

$$\rightarrow 479_{10} = 1\text{D}\text{F}_{16}$$

② Convert 1D5, 5E9, 1DF to decimal (expand by 16^n).

Convert $1D5_{16} \rightarrow$ Decimal

Hex	Decimal
1	1
D	13
5	5
256+208+1280	

→ $1D5_{16} = 1513_{10}$

Power	Calculation
$16^2 = 256$	$1 \times 256 = 256$
$16^1 = 16$	$13 \times 16 = 208$
$16^0 = 1$	$5 \times 1 = 5$

Convert $5E9_{16} \rightarrow$ Decimal

Hex	Decimal
5	5
E	14
9	9
1280+224+9=1513	

→ $5E9_{16} = 1513_{10}$

Power	Calculation
$16^2 = 256$	$5 \times 256 = 1280$
$16^1 = 16$	$14 \times 16 = 224$
$16^0 = 1$	$9 \times 1 = 9$

Convert $1DF_{16} \rightarrow$ Decimal

Hex	Decimal
1	1
D	13
F	15
1280+224+15=1519	

→ $1DF_{16} = 1519_{10}$

Power	Calculation
$16^2 = 256$	$1 \times 256 = 256$
$16^1 = 16$	$13 \times 16 = 208$
$16^0 = 1$	$15 \times 1 = 15$

3 Convert **13B4** and **A5** to **binary**

Answers: 13B4 $\rightarrow$ 0001 0011 1011 0100, A5 $\rightarrow$ 1010 0101.

Conclusion

- ✓ **Hex compresses binary while staying bit-accurate: 4 bits $\leftrightarrow$ 1 hex digit.**
- ✓ **Reading bytes as two hex digits is the de-facto standard in systems work.**
- ✓ **With base-16 powers and nibble mapping, you can convert decimal/hex/binary fluently and read real-world artifacts (colors, encodings, addresses) with confidence.**

Thank you 😊